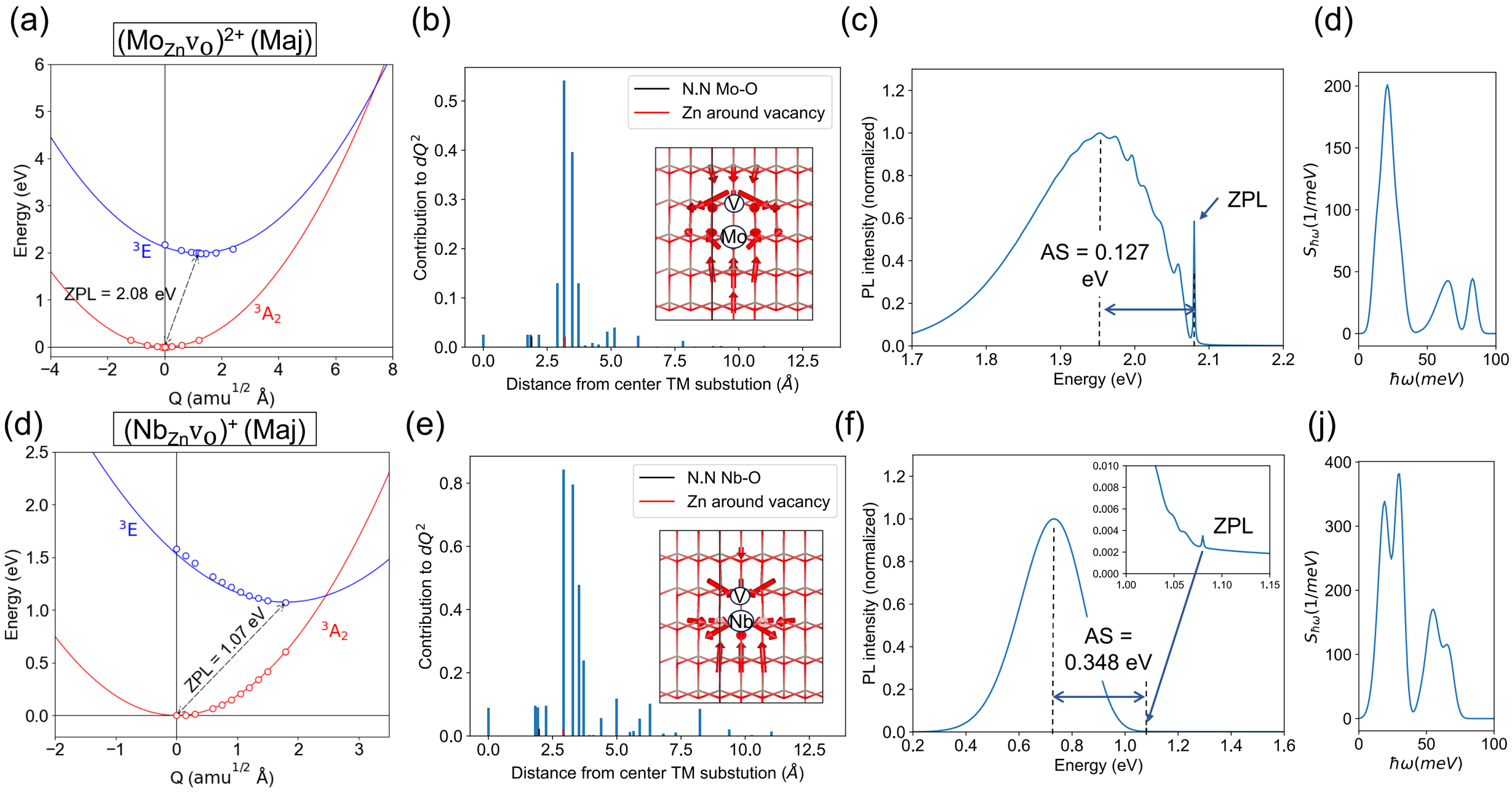
(a)
$(Mo_{Zn}v_O)^{2+}$ (Maj)
Energy (eV)
Q (amu$^{1/2}$ Å)
^3E
$^3A_2$
ZPL = 2.08 eV
(b)
N.N Mo-O
Zn around vacancy
Contribution to dQ^2
Distance from center TM substution (Å)
V
Mo
(c)
PL intensity (normalized)
Energy (eV)
ZPL
AS = 0.127 eV
(d)
$S_{\hbar\omega}(1/meV)$
$\hbar\omega(meV)$
(d)
$(Nb_{Zn}v_O)^{+}$ (Maj)
Energy (eV)
Q (amu$^{1/2}$ Å)
^3E
$^3A_2$
ZPL = 1.07 eV
(e)
N.N Nb-O
Zn around vacancy
Contribution to dQ^2
Distance from center TM substution (Å)
V
Nb
(f)
PL intensity (normalized)
Energy (eV)
ZPL
AS = 0.348 eV
(j)
$S_{\hbar\omega}(1/meV)$
$\hbar\omega(meV)$